

Firms' Costs — Homework

Grade 9 • VinSchool Grand Park

Topic: Fixed, variable, total & average costs **Questions:** 6 **Total marks:** 22

FORMULA REFERENCE SHEET

$$TC = TFC + TVC$$

$$TFC = TC - TVC$$

$$TVC = TC - TFC \text{ (or } AVC \times Q \text{)}$$

$$\text{Total Revenue (TR)} = \text{Price} \times Q$$

$$ATC = TC \div Q \text{ (or } AFC + AVC \text{)}$$

$$AFC = TFC \div Q$$

$$AVC = TVC \div Q$$

$$\text{Profit} = TR - TC$$

INSTRUCTIONS

Answer **all six questions** in your own document and submit it to your teacher.

Show your working on calculation questions — marks are awarded for working as well as the final answer.

Express monetary answers to **2 decimal places** where appropriate.

Paper 2 Exam Practice Question (1)

[4 marks]

- (a) Explain the difference between **fixed costs** and **variable costs**. **[2]**
- (b) Explain the difference between **total costs** and **unit costs**. **[2]**

Paper 2 Exam Practice Question (2)

[2 marks]

A manufacturer has total costs of production that increase from **\$5,000 to \$6,000** following an increase in its output from **200 units to 300 units**, with fixed costs at **\$2,000**.

Using the information above, calculate the **change in the manufacturer's average costs of production**. **[2]**

Paper 2 Exam Practice Question (3)**[4 marks]**

A firm produces miniature speakers with the following monthly cost structures:

Total output (speakers)	Total costs (\$)
100	\$5,000
200	\$8,000
300	\$11,000

- (a) Assuming constant unit variable costs, calculate the value of the **monthly fixed costs** for the firm. [2]
- (b) Calculate the **change in unit costs** of production if the firm changes from producing 100 units per month to 300 units per month. [2]

Paper 2 Exam Practice Question (4)**[2 marks]**

A hair salon business has **annual fixed costs of \$1.2 million**. The company has an annual output of **150,000 customers**, with a **variable cost per customer of \$15**.

Calculate the **total annual costs** for the hair salon. [2]

Paper 2 Exam Practice Question (5)**[4 marks]**

The following data refers to the costs and revenues for a toy manufacturer when operating at **2,000 units of output per month**.

Item	Costs and revenues (\$)
Price	\$15
Raw materials per unit	\$5
Overheads	\$500
Rent	\$2,000
Salaries	\$3,000

- (a) Calculate the **total cost** of producing 2,000 units of toys. [2]
- (b) Calculate the **profit** made by the toy producer if it manages to sell all of its output. [2]

Paper 2 Exam Practice Question (6)

[6 marks]

Complete the missing figures in the following table (all cost values are expressed in \$). Express your answers to **2 decimal places** where appropriate. Reproduce the table in your own document and fill in the missing values. **[6]**

Total output	TFC	TVC	TC	AFC	AVC	ATC
0	150	0		—	—	—
5		100				
10		200				
15		300				
20		400				
25		500				
30		600				
35		700				
40		800				

END OF HOMEWORK

Complete all six questions in your own document and submit it to your teacher.

Total: 22 marks